THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Assignment 3
Outline of solution

1. $X \sim Geometric(\alpha)$ and $Y \sim Geometric(\beta)$. For integer $n \geq 2$ we have

$$P(X+Y=n) = \sum_{k=1}^{n-1} P(X=k \cap Y=n-k) = \alpha\beta \sum_{k=1}^{n-1} (1-\alpha)^{k-1}(1-\beta)^{n-k-1}$$

since X and Y are dependent.

Case 1: $\alpha = \beta$. Then $P(X+Y=n) = \alpha^2 \sum_{k=1}^{n-1}(1-\alpha)^{n-2} = \alpha^2(1-\alpha)^{n-2}(n-1)$.

Case 2: $\alpha \neq \beta$. By the equality $(a-b)(a^{n-1}+a^{n-2}b+\cdots+ab^{n-2}+b^{n-1}) = a^n - b^n$,

$$\begin{aligned} P(X+Y=n) &= \alpha\beta \sum_{k=1}^{n-1} (1-\alpha)^{k-1}(1-\beta)^{n-k-1} \\ &= \alpha\beta \left(\frac{(1-\alpha)^{n-1} - (1-\beta)^{n-1}}{(1-\alpha)-(1-\beta)} \right) \\ &= \frac{\alpha\beta((1-\alpha)^{n-1} - (1-\beta)^{n-1})}{\beta - \alpha}. \end{aligned}$$

2. For $z \geq 0$,

$$\begin{aligned} F(z) &:= P(X+Y<z) \\ &= \frac{1}{2}\int_0^z \int_0^{z-x} (x+y)e^{-(x+y)}\, dy\, dx \\ &= \frac{1}{2}\int_0^z xe^{-x} \int_0^{z-x} e^{-y}\, dy\, dx + \frac{1}{2}\int_0^z e^{-x} \int_0^{z-x} ye^{-y}\, dy\, dx \\ &= \frac{1}{2}\int_0^z xe^{-x}(1-e^{-(z-x)})\, dx + \frac{1}{2}\int_0^z e^{-x}(1-e^{-(z-x)} - ze^{-(z-x)} + xe^{-(z-x)})\, dx \\ &= \frac{1}{2}\left(\int_0^z xe^{-x}\, dx - \int_0^z xe^{-z}\, dx + \int_0^z e^{-x}\, dx - \int_0^z e^{-z}\, dx - \int_0^z ze^{-z}\, dx + \int_0^z xe^{-z}\, dx \right) \\ &= \frac{1}{2}\left([-e^{-x} - xe^{-x}]_0^z + [-e^{-x}]_0^z - ze^{-z} - z^2e^{-z} \right) \\ &= 1 - e^{-z} - ze^{-z} - \frac{z^2}{2}e^{-z}. \end{aligned}$$

Thus the p.d.f. of $X+Y = F'(z) = e^{-z} - e^{-z} + ze^{-z} - ze^{-z} + \frac{z^2}{2}e^{-z} = \frac{z^2}{2}e^{-z}$.

3. (a) $f_X(x) = \int_0^x \frac{1}{x}\, dy = \frac{1}{x}[y]_0^x = 1$.

   (b) $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = f_{X,Y}(x,y) = \frac{1}{x}$.

   (c) Since we always have $Y \le X$, if $X \le \frac{1}{\sqrt{2}}$, then $X^2 + Y^2 \le 2X^2 \le 1$. Thus $P(X^2 + Y^2 \le 1|X = x) = 1$ when $X \le \frac{1}{\sqrt{2}}$.

   For $1 \ge X > \frac{1}{\sqrt{2}}$, $P(X^2 + Y^2 \le 1|X = x) = P(Y \le \sqrt{1-x^2}|X = x) = \int_0^{\sqrt{1-x^2}} \frac{1}{x}\, dy = \frac{\sqrt{1-x^2}}{x}$.

   (d)

$$\begin{aligned}
P(X^2 + Y^2 \le 1) &= \int_0^1 P(X^2 + Y^2 \le 1|X = x) f_X(x)\, dx \\
&= \int_0^{\frac{1}{\sqrt{2}}} 1\, dx + \int_{\frac{1}{\sqrt{2}}}^1 \frac{\sqrt{1-x^2}}{x}\, dx \\
&= \frac{1}{\sqrt{2}} + \int_{\pi/4}^{\pi/2} \frac{\cos^2\theta}{\sin\theta}\, d\theta \\
&= \frac{1}{\sqrt{2}} + \int_{\pi/4}^{\pi/2} \frac{d\theta}{\sin\theta} - \int_{\pi/4}^{\pi/2} \sin\theta\, d\theta \\
&= \frac{1}{\sqrt{2}} + \Big[\ln|\csc\theta - \cot\theta|\Big]_{\pi/4}^{\pi/2} + \Big[\cos\theta\Big]_{\pi/4}^{\pi/2} \\
&= \frac{1}{\sqrt{2}} + \ln(\sqrt{2}+1) - \frac{1}{\sqrt{2}} = \ln(\sqrt{2}+1).
\end{aligned}$$

4. We have

$$\begin{aligned}
\mathbb{E}(Ng(N)) &= \sum_{k=0}^{\infty} k g(k) \frac{\lambda^k e^{-\lambda}}{k!} \\
&= \lambda \sum_{k=1}^{\infty} g(k-1+1) \frac{\lambda^{k-1} e^{-\lambda}}{(k-1)!} \\
&= \lambda \sum_{k'=0}^{\infty} g(k'+1) \frac{\lambda^{k'} e^{-\lambda}}{k'!} \\
&= \lambda \mathbb{E}(g(N+1)).
\end{aligned}$$

Thus

$$\begin{aligned}
\mathbb{E}(Sg(S)) &= \mathbb{E}(\mathbb{E}(Sg(S)|N)) \\
&= \mathbb{E}\left(\mathbb{E}\left((\sum_{r=1}^{N} X_r) g(\sum_{r=1}^{N} X_r)\right)\right) \\
&= \mathbb{E}\left(\sum_{r=1}^{N} \mathbb{E}\left(X_r g(\sum_{r=1}^{N} X_r)\right)\right) \\
&= \mathbb{E}\left(N\mathbb{E}\left(X_1 g(\sum_{r=1}^{N} X_r)\right)\right) && \because X_i \text{ are i.i.d.} \\
&= \lambda\mathbb{E}\left(\mathbb{E}\left(X_1 g(\sum_{r=1}^{N+1} X_r)\right)\right) \\
&= \lambda\mathbb{E}\left(\mathbb{E}\left(X_0 g(X_0 + \sum_{r'=1}^{N} X_{r'})\right)\right) && (r' = r-1, X_i \text{ are i.i.d.}) \\
&= \lambda\mathbb{E}(X_0 g(X_0 + S)).
\end{aligned}$$

5. Case 1: X, Y discrete.

$$\begin{aligned}
\mathbb{E}(\psi(x)g(X)) &= \sum_x \left(\sum_y yP(Y=y|X=x)\right) g(x)P(X=x) \\
&= \sum_x \sum_y \frac{P(Y=y \cap X=x)}{P(X=x)} yg(x)P(X=x) \\
&= \sum_y y\left(\sum_x \frac{P(X=x \cap Y=y)}{P(Y=y)} yg(x)\right) P(Y=y) \\
&= \sum_y \mathbb{E}(yg(X)|Y=y)P(Y=y) \\
&= \mathbb{E}(Yg(X))
\end{aligned}$$

Case 2: X, Y continuous.

$$\begin{aligned}
\mathbb{E}(\psi(x)g(X)) &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} y\frac{f_{X,Y}(x,y)}{f_X(x)}\, dy\right) g(x)f_X(x)\, dx \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{f_{X,Y}(x,y)}{f_Y(y)} yg(x)f_Y(y)\, dy\, dx \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} yg(x)f_{X|Y}(x|y)\, dx\right) f_Y(y)\, dy \\
&= \int_{-\infty}^{\infty} \mathbb{E}(yg(X)|Y=y)f_Y(y)\, dy \\
&= \mathbb{E}(Yg(X))
\end{aligned}$$